

Core Java Interview Questions with Project Scenarios (1.5–2.5 Years Experience)

1. Explain the difference between `==` and `equals()` method in Java.
2. What is the purpose of the `hashCode()` method?
3. Can we override a static method in Java? Explain with an example.
4. What are wrapper classes in Java and why are they used?
5. How does autoboxing and unboxing work in Java?
6. What is the difference between `String`, `StringBuilder`, and `StringBuffer`?
7. Explain the difference between `final`, `finally`, and `finalize()`.
8. What are checked and unchecked exceptions? Give examples.
9. Explain try-with-resources in Java 7 and above.
10. What is the difference between `throw` and `throws`?
11. How does JVM memory structure work (Heap, Stack, Metaspace)?
12. Explain the concept of Garbage Collection in Java.
13. What is the use of the `transient` keyword?
14. How does synchronization work in Java?
15. Difference between process and thread.
16. Explain the concept of daemon threads.
17. What is the use of the `volatile` keyword in Java?
18. Difference between `sleep()` and `wait()` methods.
19. What is a deadlock? How can it be avoided in a multi-threaded application?
20. Explain the difference between `HashMap` and `Hashtable`.
21. What is the internal working of `HashMap` in Java?
22. What is a `ConcurrentHashMap` and how does it differ from `HashMap`?
23. Difference between `ArrayList` and `LinkedList`.
24. When would you prefer using `LinkedHashMap` over `HashMap`?
25. Explain the difference between `Comparable` and `Comparator` interfaces.
26. What is the use of `Collections.unmodifiableList()`?
27. How can you sort a list of employees by name and salary using Java 8?
28. What is a lambda expression? Provide a use case from your project.
29. How do you use `Optional` in Java 8 to avoid `NullPointerException`?
30. Explain the `map()` and `filter()` methods in Stream API.
31. What is the difference between `stream()` and `parallelStream()`?
32. What are functional interfaces? Give examples.
33. What is the difference between method reference and lambda?
34. Explain default and static methods in interfaces.
35. What is the difference between interface and abstract class?

36. Can we have private methods inside an interface in Java 9+?
37. Explain the Singleton design pattern and how to make it thread-safe.
38. What is a factory design pattern and where have you used it?
39. Explain dependency injection and its use in your project.
40. How did you handle exceptions in your project?
41. How to create custom exceptions in Java?
42. What is a POJO class?
43. Explain the use of DTO (Data Transfer Object) in your project.
44. How do you connect your Java application with a database (JDBC steps)?
45. What is PreparedStatement and how is it different from Statement?
46. What are transactions and how do you manage them in Java?
47. How to handle rollback in case of failure during database operations?
48. Explain how you implemented pagination in your project.
49. How did you implement file upload/download in your project?
50. What are some performance issues you faced and how did you optimize them?
51. Explain a real-time scenario where you used multi-threading.
52. How would you design an Employee Management System in Java?
53. How would you log all user activities in your project?
54. Explain how you can implement caching in a Java web application.
55. What is serialization? When would you use it?
56. How to make a class immutable in Java?
57. What is deep copy and shallow copy?
58. What is the use of equals() and hashCode() in HashSet?
59. How to remove duplicate elements from a list using Java 8 Streams?
60. Write code to find the second highest salary from a list of employees.
61. Write a program to find duplicate elements in an array using Java 8.
62. How to reverse a string using recursion?
63. How to check if a number is prime using Java 8 Streams?
64. Write a program to count occurrences of each word in a sentence using Streams.
65. What are the differences between fail-fast and fail-safe iterators?
66. What is serialization and how do you prevent it?
67. How do you handle concurrent modification exception?
68. Explain what happens when you call System.exit(0) in Java.
69. Explain the life cycle of a thread.
70. What is the difference between notify() and notifyAll()?
71. What is a Future and Callable in Java?
72. How to execute multiple threads in parallel using ExecutorService?
73. What is the use of synchronized keyword in Java?

74. How to prevent deadlock in multithreading?
75. What are some Java 8 improvements in collections framework?
76. Explain LocalDate, LocalTime, and LocalDateTime APIs.
77. What are the new features of Java 8 you used in your project?
78. Explain what is reflection API in Java.
79. How did you debug a production issue in your project?
80. How do you handle large data sets in your project efficiently?
81. What is difference between transient and volatile keyword?
82. Explain how you wrote unit tests for your Java classes.
83. What is the difference between static and instance variables?
84. What is a memory leak and how to identify it in Java?
85. Can you explain how equals() and hashCode() are used in collections?
86. In your project, how did you manage application configuration securely?
87. Explain a bug you fixed that involved multi-threading or synchronization.
88. How do you optimize a slow SQL query from Java code?
89. What steps do you follow before deploying your Java application?